CAFE SALES ANALYTICS CASE STUDY

End-to-end data analytics project | SWYNEX Internship — Task 4

Prepared by	Naman Shukla
Tools	Python, Pandas, exploratory data analysis, Microsoft Power BI
Dataset	Cafe Sales — 10,000 records and 8 columns in the cleaned dataset used for analysis

Project objective

Turn cafe transaction data into understandable business insights by reviewing data quality, analyzing item and revenue performance, and presenting the results in an interactive Power BI dashboard.

1. Executive Summary

This project combines data preparation, exploratory data analysis (EDA), and dashboard reporting into one analytics case study. The analysis focuses on product performance, revenue patterns, payment methods, and location records. The results also highlight data-quality limitations that should be considered before making business decisions.

Key findings from the current EDA include Coffee as the highest-volume item (3,534 units), Salad as the highest item-level revenue contributor (17,320), a positive correlation of approximately 0.704 between Quantity and Total Spent, and 3,178 transactions with an Unknown payment method.

2. Problem Statement

A cafe needs a clear view of which items sell most, which items generate the most revenue, how revenue changes over time, and how payment methods and locations are represented in transaction records. The goal is to summarize the available transaction data into useful, transparent insights while identifying data-quality issues that may affect interpretation.

Analysis questions

- Which items have the highest sales quantity and item-level revenue?
- How does revenue vary across months in the available data?
- How are transactions distributed across payment methods?
- What does location-level revenue show, and how much is affected by Unknown location records?
- What data-quality limitations should be addressed before using the results operationally?

3. Dataset Information

The cleaned dataset used in the existing EDA contains 10,000 records and 8 columns. The fields are Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date.

Field	Purpose
Transaction ID	Identifier used to count distinct transactions.
Item	Product/category used for item-level analysis.
Quantity	Number of units recorded per transaction.
Price Per Unit	Unit price field in the dataset.
Total Spent	Transaction amount used for revenue-oriented analysis.
Payment Method	Payment category; includes Unknown values.
Location	Transaction location field; includes Unknown values.
Transaction Date	Date field used for monthly trend analysis.

4. Data Cleaning & Preparation

The project uses the cleaned Cafe Sales CSV prepared in the earlier internship tasks. The final repository should include the notebook or code that makes the preparation process reproducible. In the notebook, document the exact checks performed and report the before/after row counts.

Recommended items to document and verify

- Inspect dataset shape, column names, data types, and sample records.
- Check missing, blank, and Unknown values separately.
- Check duplicate records and decide whether they are valid transactions or duplicates before removing anything.
- Convert Quantity, Price Per Unit, Total Spent, and Transaction Date to suitable data types.• Validate whether Total Spent is consistent with Quantity × Price Per Unit where applicable.
- Save the cleaned dataset and state exactly which records were changed, retained, or excluded.

Important limitation

Unknown payment and location values are meaningful data-quality limitations. Keep them visible in the analysis or clearly document any exclusion. Do not silently treat Unknown location as a real business location.

5. Exploratory Data Analysis

The earlier EDA produced the following findings. Re-run the notebook and ensure these values match the final committed dataset before submission.

Analysis	Current finding	Interpretation
Quantity by item	Coffee: 3,534 units	Coffee is the highest-volume item in the current analysis.
Revenue by item	Salad: 17,320	Salad has the highest item-level revenue in the current analysis.
Payment quality	3,178 Unknown payment transactions	Payment-method reporting is limited until these values are clarified.
Location quality	Unknown location revenue: 35,337.5	Location comparisons should distinguish unknown records from identified locations.
Correlation	Quantity vs Total Spent: approx. 0.704	A positive relationship exists; correlation alone does not establish causation.
Monthly revenue	Jan 2023: 7,242.0; Feb 2023: 6,633.5	Monthly values should be compared across the complete available date range.

6. Power BI Dashboard

The interactive dashboard developed in Task 3 is reused as the reporting layer for this final case study. It contains four KPI cards—Total Revenue, Total Transactions, Total Quantity Sold, and Average Transaction Value—and visuals for revenue by item, quantity sold by item, monthly revenue trend, transactions by payment method, and revenue by location.

Interactive slicers allow users to filter by Item, Payment Method, Location, and Transaction Date. Before submission, test each slicer and verify that the KPI cards and charts respond as expected. Ensure the monthly chart is sorted chronologically.

7. Business Insights & Recommendations

- Product planning: review Coffee demand and consider whether inventory and availability are aligned with its highunit volume.
- Revenue planning: investigate Salad’s revenue contribution alongside its quantity, pricing, and margin beforemaking promotional decisions.
- Data quality: investigate why payment method is Unknown for 3,178 transactions and improve data capture at thesource.
- Location reporting: separate Unknown location records from identified locations and improve location capture.
- Trend monitoring: use the monthly revenue chart to monitor changes over time, while checking that all dates arecorrectly typed and chronologically sorted.
- Decision quality: validate totals, date coverage, and the meaning of revenue fields before using results foroperational or financial decisions.

8. Conclusion

This case study demonstrates an end-to-end analytics workflow: preparing a transaction dataset, exploring sales patterns, identifying data-quality limitations, and presenting results through an interactive Power BI dashboard. The analysis identifies useful product-level findings while making the Unknown payment and location records visible. The next improvement would be to validate the data at source and refresh the analysis after those quality issues are resolved.

Final submission checklist

Before publishing, confirm the report matches the final CSV and notebook; add a dashboard screenshot to the report if desired; include the PDF in the Report folder; and verify that all repository links are public and open correctly.